

Total No. of Questions : 8]

SEAT No. :

PE-2181

[Total No. of Pages : 2

[6584]-80

B.E. (Civil)

AIR POLLUTION AND CONTROL

(2019 Pattern) (Semester - VII) (401004 A) (Elective - IV)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q1 or 2, Q3 or 4, Q5 or 6, Q7 or 8.
- 2) Figures to the right indicates full marks.
- 3) Draw neat figures wherever necessary.
- 4) Assume necessary data.
- 5) Use of scientific calculators is allowed.

Q1) a) Describe the process of conducting an air pollution survey and outline the key factors, both basic and statistical, to consider when choosing sampling locations? **[6]**

b) Enlist and explain the techniques and equipment employed in the sampling particulate matter. **[6]**

c) Explain isokinetic sampling? **[5]**

OR

Q2) a) Derive stack emission monitoring for particulate and gaseous matter? **[6]**

b) What does ambient air quality monitoring entail according to the CPCB, and what are the National Ambient Air Quality Standards? **[6]**

c) Convert the 50 µg/m³ of SO₂ concentration in ppm. Assume temperature 25 and pressure at 103.193 KPa. **[5]**

Q3) a) Define emission inventory and elucidate its role in air quality management? **[6]**

b) Clarify the concepts of the bottom-up and top-down approaches to developing emission inventories. **[6]**

c) Enumerate the various classifications of air quality modelling? **[5]**

OR

P.T.O.

- Q4) a)** Describe the emission inventory framework established by the CPCB? **[6]**
- b)** What are the categorizations of air quality models based on their characteristics, which include physical, statistical, and deterministic types? **[6]**
- c)** Explain assumptions, strengths, and limitations of AMREOD model USEPA? **[5]**

- Q5) a)** Explain working principle of a gravity settling chamber with neat sketch. Also, state advantages and disadvantages? **[9]**
- b)** With a neat sketch explain the principles, construction, and working of the cyclone? Also, state advantages and disadvantages? **[9]**

OR

- Q6) a)** Explain in detail working of Electrostatic Precipitator. State its advantages and disadvantages? **[9]**
- b)** Outline the criteria used to select control equipment and elaborate on their associated objectives? **[9]**

- Q7) a)** Define indoor air pollution with Sources and causes of indoor air pollution? **[6]**
- b)** Define sick building syndrome and elaborate on the recommended remedial measures to address it? **[6]**
- c)** Describe the process of measuring odours, Determination of type and intensity of odour? **[6]**

OR

- Q8) a)** Elaborate on the concept of odour pollution, covering its sources, methods of measurement, and techniques for control? **[6]**
- b)** Describe the strategies for managing indoor air pollution and provide an explanation of air cleaning systems? **[6]**
- c)** Explain in detail the techniques for managing indoor air pollution and list the plant species employed for this purpose? **[6]**
